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Theory *for* PEST Users

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1 Linear Model Theory and Terminology

Before discussing parameter estimation and uncertainty analysis for nonlinear models, we need to review linear model theory as many of the ideas and methods of estimation and analysis (inference) in nonlinear models are essentially *linear* methods applied to a linear approximate of the nonlinear models. For explanatory purposes, the parameter estimation and uncertainty analysis are discussed within the context of regression models. They can be extended to state-space or compartmental models.

1.1 A motivation example

In hydrological and water quality studies, rating curve methods in form (1) are commonly used to describe the relationship between two variables, such as stage-discharge relationship, or suspended solid concentration versus stream discharge, etc.

$$Q = bH^d \tag{1}$$

in which, Q and H are two variables whose relationship are to be determined by estimating two coefficients b and d. In order to estimate the coefficients b and d, the nonlinear form of equation (1) is usually transformed into a linear form of (2) by taking logarithm for both sides.

$$log(Q) = log(b) + d \cdot log(H) \tag{2}$$

Loosely, if we designate $y = log(Q)$, $\beta_0 = log(b)$, $\beta_1 = d$, and $x = log(H)$, then equation (2) can be rewritten as

$$y = \beta_0 + \beta_1 x. \tag{3}$$

Actually, in equation (3) y should be written as $E(y)$ representing the *mean* relation between the observations of x and y. Then, through ordinary least squares or maximum likelihood estimation method and appropriate reverse transformation, the original coefficients b and d can be estimated for equation (1).

There are two underlying points in this simple example:

1. The method of (logarithmic) transformation employed in this example is one of the linearization methods that transforms the problem of non-linear model calibration to that of linear model calibration, to which, the linear model theories can be applied. Another type of linearization method, that is, local linearization through Taylor expansion will be illustrated in Section (2).

2. Equation (3) is a simple linear regression model, by *simple* we mean that there is only one x to predict the response y. A general (multiple) linear regression model and associated methods of parameter estimation and uncertainty analysis will be presented in the following sections.

1.2 General linear regression model

In general, a *multiple linear regression* model can be expressed as

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \varepsilon, \tag{4}$$

where y is the *dependent* or *response* variable and x's are the *independent* or *predictor* variables. The β's are *parameters* or sometimes called *regression coefficients*. Thus, p is the number of the parameters in the model. The random variable ε is the *error* term in the model. In this context, error does not mean mistake but is a statistical term accounting for pure random fluctuations, measurement noise and model structural inadequacies. The designation *linear* indicates that the model (4) is linear in the parameters (i.e., the β's). For example, a model such as $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2^2 + \varepsilon$ is linear, whereas the model $y = \beta_0 + \beta_1 x_1 + e^{\beta_2 x_2} + \varepsilon$ is not a linear model.

In regression models, the explanatory variables (x_1, x_2, ..., x_p) are treated as non-random variables by assumption that they have been set to their observed values by design. That is, regression models specify the *conditional* distribution of $y|x_1, x_2, \ldots, x_p$. In particular, regression models are primarily concerned with the *mean* of this distribution, $E(y|x_1, x_2, \ldots, x_p)$, the conditional expectation of the response variable given the values of the explanatory variables, which is also known as the regression function (or expectation function). Since we always condition on the explanatory variables in regression models, we usually write $E(y)$ in place of $E(y|x_1, x_2, \ldots, x_p)$ for notational convenience.

To estimate the β's in (4), we will use a sample of n observations of y and the associated x's. The model for the ith observation is

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i, \qquad i = 1, 2, \ldots, n. \tag{5}$$

or

$$y_i = \mathbf{x}_i^T \boldsymbol{\beta} + \varepsilon_i, \qquad i = 1, 2, \ldots, n. \tag{6}$$

where $\mathbf{x}_i = (1, x_{i1}, x_{i2}, \ldots, x_{ip})^T$, and $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \ldots, \beta_p)$.

To complete the model in (5), we make the following assumptions:

1. $E(\varepsilon_i) = 0$ for all $i = 1, 2, \ldots, n$, or, equivalently, $E(y_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}$. That is to say, the model (5) is correct in representing the real system.

2. $\text{var}(\varepsilon_i) = \sigma^2$ for all $i = 1, 2, \ldots, n$, or, equivalently, $\text{var}(y_i) = \sigma^2$. In other words, the variance of ε or y does not depend on the values of x's or $E(y_i)$. This assumption is also known as the assumption of homoscedasticity or of homogenous variance or constant variance.

3. $\text{cov}(\varepsilon_i, \varepsilon_j) = 0$ for all $i \neq j$, or, equivalently, $\text{cov}(y_i, y_j) = 0$. That is, the ε's (or the y's) are uncorrelated with each other. This assumption is also known as the assumption of independence.

4. $\varepsilon_i \sim N(0, \sigma^2)$, for all $i = 1, 2, \ldots, n$. That is, the ε's are normally distributed. This assumption is also known as the assumption of normality.

When dealing with real world data sets, especially observations of flow and water quality time series in environmental sciences, any of these assumptions

may fail to hold. Normally, a plot of the data set will often reveal departures from assumptions 1 to 3. To test the assumption of normality (assumption 4), however, more sophisticated diagnostic methods such as Q-Q plot are needed. Note that the stronger assumption of normality (i.e., assumption 4) on ε is not required for the establishment of the estimation theory of linear models (that is, Gauss-Markov Theorem). But, it is necessary in order to make statistical inference (or uncertainty analysis) about parameter estimates and/or model predictions (see Section 1.4).

Writing (5) for each of the n observations, we have

$$\begin{aligned} y_1 &= \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \cdots + \beta_p x_{1p} + \varepsilon_1 \\ y_2 &= \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \cdots + \beta_p x_{2p} + \varepsilon_2 \\ &\vdots \\ y_n &= \beta_0 + \beta_1 x_{n1} + \beta_2 x_{n2} + \cdots + \beta_p x_{np} + \varepsilon_n. \end{aligned}$$

These n equations can be written in matrix from as

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ 1 & x_{21} & x_{22} & \dots & x_{2p} \\ \dots & \dots & \dots & \ddots & \dots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

or

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \tag{7}$$

Then, the above four assumptions can be expressed in terms of model (7):

1. $E(\boldsymbol{\varepsilon}) = \mathbf{0}$ or $E(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta}$.
2. $\text{cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$ or $\text{cov}(\mathbf{y}) = \sigma^2\mathbf{I}_n$.
3. $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2\mathbf{I}_n)$.

Note that assumption 3 subsumes the second assumption and all three assumptions can be succinctly expressed as $\mathbf{y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I}_n)$.

Before we leave this section, please note that in the linear model the (i, j)th element in the matrix $\mathbf{X}$ is

$$x_{ij} = \frac{\partial(y_i)}{\partial\beta_j} = \frac{\partial(\mathbf{x}_i^T\boldsymbol{\beta})}{\partial\beta_j}.$$

Or, it is not difficult to see that the matrix $\mathbf{X}$ is the derivative of $\mathbf{X}\boldsymbol{\beta}$ with respect to $\boldsymbol{\beta}$, that is,

$$\mathbf{X} = \frac{\partial(\mathbf{X}\boldsymbol{\beta})}{\partial \boldsymbol{\beta}^T}. \tag{8}$$

For this reason, $\mathbf{X}$ is called the *derivative matrix* (it is also called the design matrix). In linear regression models, the derivative matrix $\mathbf{X}$ does not depend on the model parameters $\boldsymbol{\beta}$. However, this will not be the case in nonlinear models (see Equation (27)).

1.3 Parameter estimation

The problem of estimation is to obtain the "best" values ($\hat{\boldsymbol{\beta}}$) for the parameters ($\boldsymbol{\beta}$), which minimize the discrepancy between the model output ($E(\mathbf{y}) = \mathbf{X}\hat{\boldsymbol{\beta}}$) and the observations ($\mathbf{y}$). When the assumptions for model (7), especially assumption 1 and 2, listed in the previous section hold, the *ordinary least squares* (OLS) method can be used to estimate the model parameters ($\boldsymbol{\beta}$). However, when either the assumption of homoscedasticity or the assumption of independence (i.e., assumption 2) is violated, the *weighted least squares* (WLS) is commonly used.

1.3.1 Ordinary least squares estimator

The ordinary least squares criterion is to minimize the sum of squared errors, that is, to minimize

$$\Phi(\boldsymbol{\beta}) \equiv \sum_{i=1}^{n} \varepsilon_i^2 = \sum_{i=1}^{n} (y_i - \mathbf{x}_i^T \boldsymbol{\beta})^2 \tag{9}$$

In order to find the optimal parameter vector $\boldsymbol{\beta}$ that minimizes the *objective function* $\Phi(\boldsymbol{\beta})$, we differentiate $\Phi(\boldsymbol{\beta})$ with respect to $\boldsymbol{\beta}$ and set the result equal to zero.

We rewrite $\Phi(\boldsymbol{\beta})$ as follows:

$$\Phi(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T \mathbf{I}_n (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{y}^T\mathbf{y} - 2\mathbf{y}^T\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta}$$

Using matrix differentiation[1], we get

$$\frac{\partial \Phi(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}^T} = -2\mathbf{y}^T\mathbf{X} + 2\boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}.$$

[1]For vector $\mathbf{x}$ and matrix $\mathbf{A}$, $\frac{\partial \mathbf{A}\mathbf{x}}{\partial \mathbf{x}^T} = \mathbf{A}$; and $\frac{\partial \mathbf{x}^T\mathbf{A}\mathbf{x}}{\partial \mathbf{x}^T} = 2\mathbf{x}^T\mathbf{A}$.

Therefore, the *normal equations* are

$$[\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}]^T\mathbf{X} = \mathbf{0}.$$

or,

$$\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y}. \tag{10}$$

If $\mathbf{X}^T\mathbf{X}$ is nonsingular, we can multiply both sides of the normal equations by $(\mathbf{X}^T\mathbf{X})^{-1}$ to give the ordinary least squares estimator of $\boldsymbol{\beta}$ as:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} \tag{11}$$

which minimizes the objective function defined in (9).

1.3.2 Weighted least squares estimator

In environmental modeling, the observations of environmental parameters, such as stream discharge, are normally not independent from each other since they are time series observations. In addition, these observations have heterogenous variances. For example, the measurement errors for high flows are normally larger than those of low flows. These violate the assumptions of homoscedasticity or constant variances of the errors (assumption 2). This means that the variance-covariance matrix of the error term takes the form:

$$\text{var}(\boldsymbol{\varepsilon}) = \sigma^2\boldsymbol{\Sigma} \tag{12}$$

where $\boldsymbol{\Sigma}$ is a *known* positive-definite matrix and not necessarily equal to $\mathbf{I}_n$ as required in ordinary least squares.

Then we multiply both sides of (7) by $\boldsymbol{\Sigma}^{-\frac{T}{2}} \equiv \{(\boldsymbol{\Sigma}^{1/2})^T\}^{-1}$ to obtain an equivalent transformed model,

$$\boldsymbol{\Sigma}^{-\frac{T}{2}}\mathbf{y} = \boldsymbol{\Sigma}^{-\frac{T}{2}}\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\Sigma}^{-\frac{T}{2}}\boldsymbol{\varepsilon}$$

That is,

$$\mathbf{y}^* = \mathbf{X}^*\boldsymbol{\beta} + \boldsymbol{\varepsilon}^* \tag{13}$$

Since the error term $\boldsymbol{\varepsilon}^*$ in model (13) fulfills the assumptions of homoscedasticity or constant variances, the ordinary least squares estimator can now be applied to obtain the estimate of $\boldsymbol{\beta}$ in model (13). That is

$$\hat{\boldsymbol{\beta}} = \{(\mathbf{X}^*)^T\mathbf{X}^*\}^{-1}(\mathbf{X}^*)^T\mathbf{y}^*$$

Substituting $\mathbf{X}^* = \mathbf{\Sigma}^{-\frac{T}{2}}\mathbf{X}$ and $\mathbf{y}^* = \mathbf{\Sigma}^{-\frac{T}{2}}\mathbf{y}$, we have

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{\Sigma}^{-1}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{\Sigma}^{-1}\mathbf{y} \tag{14}$$

It is not difficult to show that $\hat{\boldsymbol{\beta}}$ obtained from equation (14) minimizes the objective function

$$\Phi(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T\mathbf{Q}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \tag{15}$$

where $\mathbf{Q}$ is equal to $\mathbf{\Sigma}^{-1}$ and called the *cofactor matrix* in the PEST manual. According to equation (12), this cofactor matrix is proportional to the observation covariance matrix var($\boldsymbol{\varepsilon}$) with the constant of proportionality being σ^2, which is called the *reference variance* in the PEST manual. Equation (14) then is written in terms of this cofactor matrix as:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{Q}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Q}\mathbf{y} \tag{16}$$

If only the assumption of homoscedasticity is violated, then $\mathbf{\Sigma}$ in equation (12) simply contains diagonal elements, so does the matrix $\mathbf{Q}$ in (15), with its elements being the reciprocals of the corresponding elements in $\mathbf{\Sigma}$. Therefore, the assumption of independence of the observations is maintained through requiring that $\mathbf{Q}$ only have diagonal elements, with their values being the squares of the observation weights w_i's. That is,

$$\mathbf{Q} = \mathbf{\Sigma}^{-1} = \begin{pmatrix} w_1^2 & 0 & \dots & 0 \\ 0 & w_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & w_n^2 \end{pmatrix} \tag{17}$$

Therefore, the objective function (15) that the WLS estimates minimize can be rewritten as

$$\Phi(\boldsymbol{\beta}) = \sum_{i=1}^{n}[w_i(y_i - \mathbf{x}_i^T\boldsymbol{\beta})]^2$$

When PEST is used to automatically calibrate a model, in order to use the WLS method, one may either specify a weight for each of the observations which model outputs are to be calibrated against, or supply an observation covariance matrix. The weights are supplied to rectify the violation of constant variances only, while the covariance matrix with non-zero off-diagonal elements is provided to mitigate the violations of homoscedasticity and independence.

1.4 Uncertainty analysis

The uncertainty analysis associated with linear regression models provides the confidence intervals for model parameters (the β's) and the *mean* responses of the model ($E(y)$), and *prediction* intervals for a future observation. As mentioned earlier, in order to proceed with uncertainty analysis, the assumption of normality on observations (or errors) is mandated. However, in order to obtain more general results and be consistent with the PEST manual, the assumptions of homoscedasticity and independence (assumption 2) are hereby relaxed. In other words, the uncertainty analysis is applied to the results of the WLS estimation instead of the OLS estimation. However, it should be an easy matter to transfer these results to the latter (namely, by changing $\mathbf{\Sigma}$ into $\mathbf{I}_n$ wherever it is applicable). Therefore, throughout this section, it is assumed that $\mathbf{y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{\Sigma})$.

In this section, some of the final uncertainty analysis results are presented with their derivation being omitted. However, the detailed proofs for these results can be readily found in any textbooks of linear regression models.

1.4.1 Variance-covariance matrix of $\hat{\boldsymbol{\beta}}$ and estimation of σ^2

Following the fact that $\hat{\boldsymbol{\beta}}$ is a *linear* function of $\mathbf{y}$, it is easy to find that $\hat{\boldsymbol{\beta}}$, the estimate of $\boldsymbol{\beta}$ from the WLS method, has a variance-covariance matrix

$$\mathrm{var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^T\mathbf{\Sigma}^{-1}\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^T\mathbf{Q}\mathbf{X})^{-1} \tag{18}$$

and (based on the assumption of normality)

$$\hat{\boldsymbol{\beta}} \sim N_p\big(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^T\mathbf{Q}\mathbf{X})^{-1}\big). \tag{19}$$

Since σ^2 is unknown, it is usually estimated by the mean squared error s^2 as follows,

$$\hat{\sigma}^2 = s^2 = \frac{\Phi(\boldsymbol{\beta})}{n-p} = \frac{\sum_{i=1}^{n}[w_i(y_i - \mathbf{x}_i^T\boldsymbol{\beta})]^2}{n-p} \tag{20}$$

where n is the number of observations and p is the number of parameters to be estimated. Thus the degrees of freedom are $n - p$.

1.4.2 Confidence interval for β_j

From equations (18) and (20), we have

$$\widehat{\mathrm{var}}(\hat{\beta}_j) = s^2(\mathbf{X}^T\mathbf{Q}\mathbf{X})^{-1}_{jj}$$

where subscript jj indicates the jth diagonal element of $(\mathbf{X}^T\mathbf{Q}\mathbf{X})^{-1}$. The square root of this estimated variance is the *standard error* of $\hat{\beta}_j$, That is,

$$\text{s.e.}(\hat{\beta}_j) = s\sqrt{g_{jj}}$$

where g_{jj} is the jth diagonal element of $(\mathbf{X}^T\mathbf{Q}\mathbf{X})^{-1}$. Then, it is shown that the statistic

$$t_j = \frac{\hat{\beta}_j - \beta_j}{\text{s.e.}(\hat{\beta}_j)}$$

has the central t-distribution with $n-p$ degrees of freedom. Then,

$$P\bigg[-t_{\alpha/2}(n-p) \leq \frac{\hat{\beta}_j - \beta_j}{\text{s.e.}(\hat{\beta}_j)} \leq t_{\alpha/2}(n-p)\bigg] = 1-\alpha$$

Therefore, the $100(1-\alpha)\%$ confidence interval for β_j is

$$\hat{\beta}_j \pm t_{\alpha/2}(n-p)\text{s.e.}(\hat{\beta}_j) \tag{21}$$

1.4.3 Confidence region for $\boldsymbol{\beta}$

Note that the confidence interval obtained above is for the individual β's. However, these parameters β_0, β_1, β_2, …, β_p in $\boldsymbol{\beta}$ vary in harmony instead of separately, therefore, we should also consider their joint confidence region. It can be shown that a $100(1-\alpha)\%$ joint confidence region for $\boldsymbol{\beta}$ is given by all vectors $\boldsymbol{\beta}$ that satisfy

$$(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^T\mathbf{X}^T\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \leq ps^2F_\alpha(p, n-p) \tag{22}$$

where $F_\alpha(p, n-p)$ is F-distribution with $(p, n-p)$ degrees of freedom. For $p = 2$, this region can be plotted as an ellipse in two dimensions as shown in Figure 1. For $p > 2$, the elliptical region (ellipsoid) in (22) is unwieldy to illustrate.

1.4.4 Confidence interval for $E(y_0)$

Let $\mathbf{x}_0 = (1, x_{01}, x_{02}, \ldots, x_{0p})^T$ denote a explanatory variable vector of a future event and y_0 be an observation corresponding to $\mathbf{x}_0$. Then

$$y_0 = \mathbf{x}_0^T\boldsymbol{\beta} + \varepsilon,$$

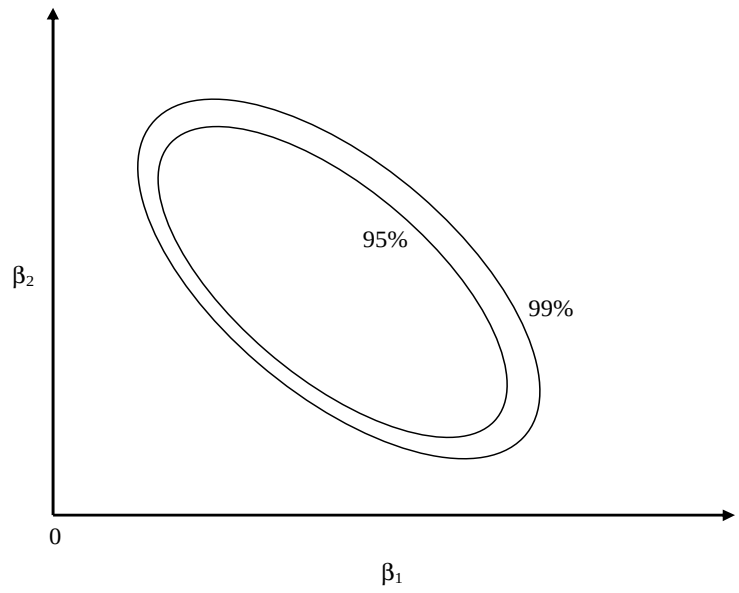


Figure 1: Confidence region for $\boldsymbol{\beta} = (\beta_1, \beta_2)^T$.

and,

$$E(y_0) = \mathbf{x}_0^T \boldsymbol{\beta}.$$

Using simple matrix relationships for propagation of covariance, the standard error of $\widehat{E(y_0)}$ (the estimator of $E(y_0)$), denoted as s.e.$(\hat{y}_0)$ is easily calculated as

$$\text{s.e.}(\hat{y}_0) = s\sqrt{\mathbf{x}_0^T(\mathbf{X}^T\mathbf{Q}\mathbf{X})^{-1}\mathbf{x}_0}$$

Then, the confidence interval for $E(y_0)$, the *mean* of the distribution of y-value corresponding to $\mathbf{x}_0$ is

$$\mathbf{x}_0^T\hat{\boldsymbol{\beta}} \pm t_{\alpha/2}(n-p)\text{s.e.}(\hat{y}_0) \tag{23}$$

1.4.5 Prediction interval for a future observation y_0

A "confidence interval" for a future observation y_0 corresponding to $\mathbf{x}_0$ is called a *prediction* interval. We speak of a prediction interval rather than a confidence interval because y_0 is an *individual* observation and is thereby a random variable rather than a *mean* ($E(y_0)$ of the samples of the random

variable. Therefore, at the same level of significance, the prediction interval for y_0 will clearly have to be wider than a confidence interval for $E(y_0)$.

Since the standard error

$$\text{s.e.}(y_0 - \hat{y}_0) = s\sqrt{1 + \mathbf{x}_0^T(\mathbf{X}^T\mathbf{Q}\mathbf{X})^{-1}\mathbf{x}_0}$$

and the t-statistic

$$t = \frac{(y_0 - \hat{y}_0) - 0}{\text{s.e.}(y_0 - \hat{y}_0)}$$

is distributed as $t(n-p)$, then we have the $100(1-\alpha)\%$ prediction interval for y_0 as

$$\mathbf{x}_0^T\hat{\boldsymbol{\beta}} \pm t_{\alpha/2}(n-p)\text{s.e.}(y_0) \tag{24}$$

where $\text{s.e.}(y_0) = s\sqrt{1 + \mathbf{x}_0^T(\mathbf{X}^T\mathbf{Q}\mathbf{X})^{-1}\mathbf{x}_0}$, in which, the second term, $\mathbf{x}_0^T(\mathbf{X}^T\mathbf{Q}\mathbf{X})^{-1}\mathbf{x}_0$, is typically much smaller than 1 provided p is much smaller than n.

2 Nonlinear regression model

Now suppose we have a nonlinear model taking the following form

$$y_i = f(\mathbf{x}_i, \boldsymbol{\theta}) + \varepsilon_i \tag{25}$$

with y_i, $\mathbf{x}_i$ and ε_i defined as in the linear model (6). But the *expectation* (or regression) function $f(\cdot)$ is nonlinear in any component of $\boldsymbol{\theta}$ (note that we have changed the notation for model parameter). That is, $\frac{\partial f(\mathbf{x},\boldsymbol{\theta})}{\partial \theta_j}$ depends on $\boldsymbol{\theta}$ for some j as contrasted in the linear model, in which the derivative matrix is independent from parameters (see equation (8)). If letting $\eta_i(\boldsymbol{\theta}) = f(\mathbf{x}_i, \boldsymbol{\theta})$ and $\boldsymbol{\eta} = (\eta_1, \ldots, \eta_n)^T$, then we can write the nonlinear model for n observations as

$$\mathbf{y} = \boldsymbol{\eta}(\boldsymbol{\theta}) + \boldsymbol{\varepsilon} \tag{26}$$

in which, we assume $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.

2.1 Linear approximation

Estimation about $\boldsymbol{\theta}$ is much easier if $f(\mathbf{x}_i, \boldsymbol{\theta})$ is linear in $\boldsymbol{\theta}$ as we described in the previous section. This suggests using a linear Taylor series approximation of $f(\mathbf{x}_i, \boldsymbol{\theta})$. For $\boldsymbol{\theta}$ near $\boldsymbol{\theta}^*$,

$$f(\mathbf{x}_i, \boldsymbol{\theta}) \approx f(\mathbf{x}_i, \boldsymbol{\theta}^*) + (\theta_1 - \theta_1^*)J_{i1} + \cdots + (\theta_p - \theta_p^*)J_{ip}$$

for each $i = 1, \ldots, n$, where

$$J_{ij} = \left.\frac{\partial f(\mathbf{x}_i, \boldsymbol{\theta})}{\partial \theta_j}\right|_{\boldsymbol{\theta}=\boldsymbol{\theta}^*} \tag{27}$$

which is the first derivative of the ith observation with respect to the jth parameter in $\boldsymbol{\theta}$. Then, stacking these n approximations on top of one another in vector form, we will have a linear approximation for the model of (26)

$$\mathbf{y} \approx \boldsymbol{\eta}(\boldsymbol{\theta}^*) + \mathbf{J}(\boldsymbol{\theta}^*)(\boldsymbol{\theta} - \boldsymbol{\theta}^*) + \boldsymbol{\varepsilon} \tag{28}$$

where $\mathbf{J}(\boldsymbol{\theta}^*)$ is the $n \times p$ matrix with (i, j)th element being J_{ij}. Therefore, $\mathbf{J}(\boldsymbol{\theta}^*)$ corresponds to the derivative matrix $\mathbf{X}$ in the linear model (see Equation (8)), but it is commonly called *Jacobian matrix*.

2.2 Nonlinear least squares estimator

As in the OLS, the nonlinear least squares estimator is to find the optimal value for $\boldsymbol{\theta}$ that minimizes the sum of squared errors, i.e., the objective function:

$$\Phi(\boldsymbol{\theta}) = \|\mathbf{y} - \boldsymbol{\eta}(\boldsymbol{\theta})\|^2 = \sum_{i=1}^{n} \{y_i - f(\mathbf{x}_i, \boldsymbol{\theta})\}^2 \tag{29}$$

Note that the only difference from the OLS is that $f(\mathbf{x}_i, \boldsymbol{\theta}) \neq \mathbf{x}_i^T \boldsymbol{\theta}$ (nonlinearity). As in the linear case, in order to find the optimal parameter vector $\boldsymbol{\theta}$ that minimizes the *objective function* $\Phi(\boldsymbol{\theta})$, we differentiate $\Phi(\boldsymbol{\theta})$ with respect to $\boldsymbol{\theta}$ and set the result equal to zero. Taking derivatives, we have

$$\frac{\partial \Phi(\boldsymbol{\theta})}{\partial \theta_j} = -2 \sum_{i=1}^{n} \{y_i - f(\mathbf{x}_i, \boldsymbol{\theta})\} \frac{\partial f(\mathbf{x}_i, \boldsymbol{\theta})}{\partial \theta_j} \qquad j = 1, \ldots, p.$$

Or, in matrix notation,

$$\frac{\partial \Phi(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}^T} = -2[\mathbf{y} - \boldsymbol{\eta}(\boldsymbol{\theta})]^T \mathbf{J}(\boldsymbol{\theta}) \tag{30}$$

Therefore, the *normal equations* for the model (28) are

$$[\mathbf{y} - \boldsymbol{\eta}(\hat{\boldsymbol{\theta}})]^T \mathbf{J}(\hat{\boldsymbol{\theta}}) = \mathbf{0} \tag{31}$$

Recall that in linear models, the derivative matrix was $\mathbf{J}(\boldsymbol{\theta}) = \mathbf{X}$ and the normal equations took the following form:

$$[\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}]^T \mathbf{X} = \mathbf{0}$$

But, in the nonlinear case, the normal equations (31) are nonlinear in $\boldsymbol{\theta}$ and do not, in general, yield a closed form expression for the solution $\hat{\boldsymbol{\theta}}$, the NLS estimator. Usually it requires iterative numerical methods, some of which are illustrated in the coming section.

2.3 Numerical methods

All the numerical methods that are used to find the solution to the normal equations in (31) can also be treated as the *classical approaches* that find the minimum of the objective function (29) *directly*, because the coordinates

of the minimum will define the NLS estimates of the model's parameters (namely, the solution to the normal equation (31)). All these methods are *local gradient* methods. The designation *local* means that these methods do not guarantee to find the parameter set ($\boldsymbol{\theta}$) that gives the minimum of the *entire* objective function surface. In contrast, they usually converge to a local minimum of the objective function, especially when the objective function surface is very complex and irregular. The designation *gradient* means that these methods utilize information about the slope of the objective function to dictate the direction of search towards where the minimum is thought to lie. Therefore these methods are generally more efficient when the objective function is continuous in its first (and second) derivatives. When used to calibrate models, they require substantially less computer run-time than global methods.

Most gradient search strategies are based on the following generic iterative update formula to calculate $\hat{\boldsymbol{\theta}}$:

$$\hat{\boldsymbol{\theta}}^{j+1} = \hat{\boldsymbol{\theta}}^{j} + \boldsymbol{\delta}^{j} \tag{32}$$

in which, $\hat{\boldsymbol{\theta}}^{j+1}$ is the updated parameter set in the parameter hyperspace; $\hat{\boldsymbol{\theta}}^{j}$ is the current parameter set; $\boldsymbol{\delta}^{j}$, called the *update vector* in the PEST Manual, is the jth step increment for the parameter vector, which usually consists of 3 components:

i. the length of step, ρ;

ii. a specifically chosen square matrix, $\boldsymbol{\Gamma}$; and

iii. the gradient matrix, $-\frac{\partial \Phi(\hat{\boldsymbol{\theta}}^{j})}{\partial \boldsymbol{\theta}}$, which defines the steepest downward gradient of the objective function surface, evaluated locally at the current set of parameter estimates.

Therefore the above equation can also be expressed as

$$\hat{\boldsymbol{\theta}}^{j+1} = \hat{\boldsymbol{\theta}}^{j} + \rho\boldsymbol{\Gamma}\left(- \frac{\partial \Phi(\hat{\boldsymbol{\theta}}^{j})}{\partial \boldsymbol{\theta}} \right) \tag{33}$$

The objective function value $\Phi(\boldsymbol{\theta}^{j+1})$ determined by the newly updated parameter set $\boldsymbol{\theta}^{j+1}$ is not guaranteed smaller than $\Phi(\boldsymbol{\theta}^{j})$ by the iterative method in (33). Therefore, if $\Phi(\boldsymbol{\theta}^{j+1})$ is greater than $\Phi(\boldsymbol{\theta}^{j})$, the step size ρ

is reduced and a new parameter set is evaluated, and so on. At a specific (current) point ($\boldsymbol{\theta}^j$), the steepest downward gradient $-\frac{\partial\Phi(\hat{\boldsymbol{\theta}}^j)}{\partial\boldsymbol{\theta}}$ is fixed, therefore, the various numerical methods illustrated below only differ from each other by adopting different strategies of choosing the matrix $\boldsymbol{\Gamma}$, which can be thought as a modifier that shifts the searching direction at each step off the steepest downward direction.

2.3.1 Steepest Descent algorithm

Since the direction being pointed by $-\frac{\partial}{\partial\boldsymbol{\theta}}\Phi(\hat{\boldsymbol{\theta}}^j)$ is the most promising direction to locate the bottom-most point in the surface, it seems heuristic to let $\boldsymbol{\Gamma} = \mathbf{I}$. That is, the searching is always along with the steepest downward direction. Therefore, the steepest descent update formula takes the form

$$\hat{\boldsymbol{\theta}}^{j+1} = \hat{\boldsymbol{\theta}}^j - \rho \cdot \mathbf{I} \cdot \frac{\partial}{\partial\boldsymbol{\theta}}\Phi(\hat{\boldsymbol{\theta}}^j) \tag{34}$$

or (substituting Equation (30))

$$\hat{\boldsymbol{\theta}}^{j+1} = \hat{\boldsymbol{\theta}}^j + \rho \cdot \mathbf{I} \cdot \{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}^T\{\mathbf{y} - \boldsymbol{\eta}(\hat{\boldsymbol{\theta}}^j)\}^2 \tag{35}$$

where ρ is chosen to minimize $\Phi(\hat{\boldsymbol{\theta}}^{j+1})$ as described above.

2.3.2 Gauss-Newton algorithm

To find the numerical solution to the normal equation (31), the Gauss-Newton method proceeds through the following iterative steps:

1. Setting initial values for $\hat{\boldsymbol{\theta}}$, e.g., $\hat{\boldsymbol{\theta}}^0$;
2. Using a linear approximation to $\boldsymbol{\eta}(\boldsymbol{\theta})$ for $\boldsymbol{\theta}$ near $\hat{\boldsymbol{\theta}}^0$ (see Equation (28));
3. Using the methods for linear models to "estimate" $\boldsymbol{\theta}$, i.e., to update $\hat{\boldsymbol{\theta}}^0$ to $\hat{\boldsymbol{\theta}}^1$;
4. Repeating steps 2 and 3 until convergence.

[2] A factor of 2 is dropped out since it can be subsumed in ρ.

Let $\mathbf{J}^0 = \mathbf{J}(\hat{\boldsymbol{\theta}}^0)$. Then, for $\boldsymbol{\theta}$ near $\hat{\boldsymbol{\theta}}^0$,

$$\boldsymbol{\eta}(\boldsymbol{\theta}) \approx \boldsymbol{\eta}(\hat{\boldsymbol{\theta}}^0) + \mathbf{J}^0(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}^0)$$

It follows that

$$\begin{aligned} \mathbf{y} - \boldsymbol{\eta}(\boldsymbol{\theta}) &\approx \mathbf{y} - \boldsymbol{\eta}(\hat{\boldsymbol{\theta}}^0) - \mathbf{J}^0(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}^0) \\ &\equiv \mathbf{z}^0 - \mathbf{J}^0\boldsymbol{\delta} \end{aligned}$$

where $\mathbf{z}^0 = \mathbf{y} - \boldsymbol{\eta}(\hat{\boldsymbol{\theta}}^0)$ and $\boldsymbol{\delta} = \boldsymbol{\theta} - \hat{\boldsymbol{\theta}}^0$. Therefore, for $\boldsymbol{\theta}$ near $\hat{\boldsymbol{\theta}}^0$, the problem of choosing $\boldsymbol{\theta}$ to minimize $\|\mathbf{y} - \boldsymbol{\eta}(\boldsymbol{\theta})\|^2$ is "approximately equivalent to" the problem of choosing $\boldsymbol{\delta}$ to minimize $\|\mathbf{z}^0 - \mathbf{J}^0\boldsymbol{\delta}\|^2$. Furthermore, this minimization can be carried out using (ordinary) least squares method. For example, we could obtain

$$\hat{\boldsymbol{\delta}} = \{(\mathbf{J}^0)^T\mathbf{J}^0\}^{-1}(\mathbf{J}^0)^T\mathbf{z}^0 \tag{36}$$

similarly to equation (11). Once we have $\hat{\boldsymbol{\delta}}$, we can easily update $\hat{\boldsymbol{\theta}}^0$ to $\hat{\boldsymbol{\theta}}^1$ via

$$\hat{\boldsymbol{\theta}}^1 = \hat{\boldsymbol{\theta}}^0 + \hat{\boldsymbol{\delta}} \tag{37}$$

$\hat{\boldsymbol{\theta}}$ is updated this way until convergence. In generalization, at jth iteration, the Gauss-Newton updating formula is:

$$\hat{\boldsymbol{\theta}}^{j+1} = \hat{\boldsymbol{\theta}}^j + \left[\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}^T\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}\right]^{-1}\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}^T\{\mathbf{y} - \boldsymbol{\eta}(\hat{\boldsymbol{\theta}}^j)\} \tag{38}$$

However, there is no guarantee that the update will reduce the objective function $\Phi(\boldsymbol{\theta})$. That is to say, no guarantee that $\Phi(\hat{\boldsymbol{\theta}}^{j+1}) < \Phi(\hat{\boldsymbol{\theta}}^j)$. Fortunately, this problem can be fixed through using the step length ρ as briefly described above. If $\Phi(\hat{\boldsymbol{\theta}}^{j+1}) > \Phi(\hat{\boldsymbol{\theta}}^j)$, we replace the updating formula (38) by

$$\hat{\boldsymbol{\theta}}^{j+1} = \hat{\boldsymbol{\theta}}^j + \rho \cdot \left[\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}^T\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}\right]^{-1}\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}^T\{\mathbf{y} - \boldsymbol{\eta}(\hat{\boldsymbol{\theta}}^j)\} \tag{39}$$

where $0 < \rho \leq 1$. To choose ρ, we start with $\rho = 1$ and then try ρ equal to $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots$ until we find a ρ that gives $\Phi(\hat{\boldsymbol{\theta}}^{j+1}) < \Phi(\hat{\boldsymbol{\theta}}^j)$. Comparing formulas (33) and (39), it is not difficult to find that, in the G-N method, $\boldsymbol{\Gamma} = \left[\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}^T\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}\right]^{-1}$.

2.3.3 Levenberg-Marquardt algorithm

When the objective function has a regular response function, experiences show that the S-D method is a efficient method and relatively insensitive to starting values. However, in the situation where the objective function has a narrow valley-like response surface, probably due to highly correlated parameters, the S-D method may lead to a phenomenon called "hemstitching" (see Figure 2) where the parameter set jumps from side to side of the valley as parameters are upgraded on successive iterations. Inevitably convergence toward the global minimum is then extremely slow.

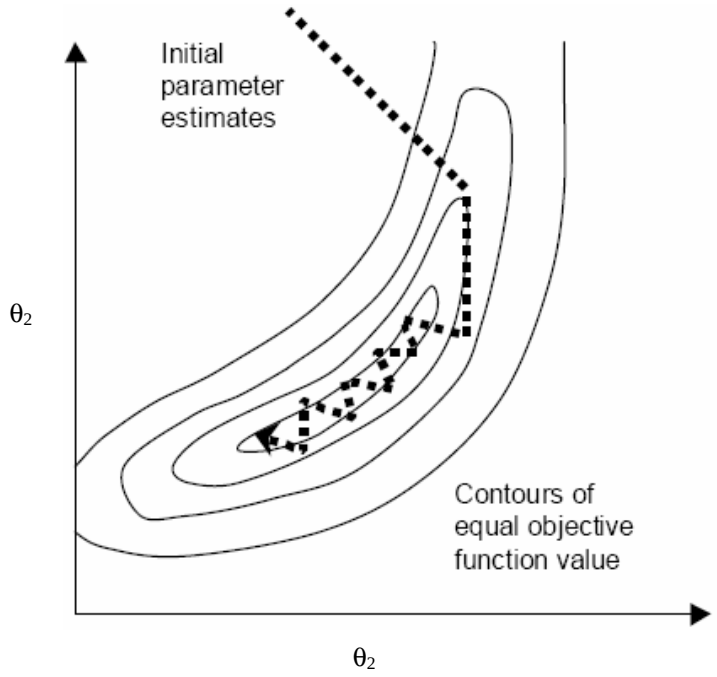


Figure 2: The phenomenon of "hemstitching" (taken from the *PEST User's Manual*).

As we've witnessed in Section 2.3.2, the G-N method changes the searching direction at each step from $-\frac{\partial\Phi(\hat{\boldsymbol{\theta}}^j)}{\partial\boldsymbol{\theta}}$ to $-\left[\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}^T\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}\right]^{-1}\frac{\partial\Phi(\hat{\boldsymbol{\theta}}^j)}{\partial\boldsymbol{\theta}}$. It will perform better than the S-D method when searching in a valley-like response surface, but a new problem may arise. If $\{\mathbf{J}(\boldsymbol{\theta})\}^T\{\mathbf{J}(\boldsymbol{\theta})\}$ is close to singular for $\boldsymbol{\theta}$ near $\hat{\boldsymbol{\theta}}$, the G-N method may become erratic. One helpful

approach to mitigating this problem is to modify the square matrix $\mathbf{\Gamma}$ to

$$\mathbf{\Gamma}(\lambda) = \left[\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}^T\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\} + \lambda\mathbf{B}\right]^{-1} \tag{40}$$

where $\mathbf{B}$ is a diagonal matrix and usually has two choices:

i. $\mathbf{B} = \mathbf{I}$ (PEST uses); and

ii. $\mathbf{B}$'s diagonal elements taken from the diagonal elements of $\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}^T\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}$.

Therefore, the update formula of the L-M method is the following:

$$\boldsymbol{\theta}^{j+1} = \boldsymbol{\theta}^j + \rho \cdot \left[\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}^T\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\} + \lambda\mathbf{B}\right]^{-1} \cdot \{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}^T\{\mathbf{y} - \boldsymbol{\eta}(\hat{\boldsymbol{\theta}}^j)\} \tag{41}$$

To choose λ, the following strategy, suggested by Marquardt (1963), is often used:

1. Start with small λ;
2. If $\Phi(\hat{\boldsymbol{\theta}}^{j+1}) < \Phi(\hat{\boldsymbol{\theta}}^j)$, then decrease λ on next iteration;
3. If $\Phi(\hat{\boldsymbol{\theta}}^{j+1}) \geq \Phi(\hat{\boldsymbol{\theta}}^j)$, then increase λ until (2) occurs.

For well-behaved problems, this approach will lead to $\lambda \to 0$ and the L-M method will behave like the G-N method. But for ill-conditioned problems, λ will increase toward ∞ and the L-M method will become a variant of the steepest descent method. Therefore, the L-M method is a compromise between the G-N and steepest descent methods. This is the method implemented by the PEST software with some modifications to achieve the best efficiency.

2.3.4 Newton's methods

It is not essential for the readers to understand the Newton's methods in order to use PEST, the model calibration tool. But they are still included in this manuscript merely for completeness. In general, Newton's methods utilize not only the first derivative but also the second derivatives of the objective function, by which the Newton's methods differ from the other methods we have so far discussed. The Newton-Raphson and the Quasi-Newton algorithms are the most commonly implemented Newton's methods in commercial optimization software.

1. *Newton-Raphson algorithm.* As discussed in Section 2.2, in the NLS, $\hat{\boldsymbol{\theta}}$ satisfies the normal equation

$$\frac{\partial \Phi}{\partial \boldsymbol{\theta}}(\hat{\boldsymbol{\theta}}) = 0$$

where $\frac{\partial \Phi}{\partial \boldsymbol{\theta}}(\hat{\boldsymbol{\theta}}) \equiv \frac{\partial \Phi(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\big|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}}$.

Now, we use a Taylor series to linearize $\frac{\partial \Phi}{\partial \boldsymbol{\theta}}(\hat{\boldsymbol{\theta}})$ (the first derivative of the objective function) instead of $\boldsymbol{\eta}(\boldsymbol{\theta})$ (the model itself) as we do in the Gauss-Newton method. We have

$$\mathbf{0} = \frac{\partial \Phi}{\partial \boldsymbol{\theta}}(\hat{\boldsymbol{\theta}}) \approx \frac{\partial \Phi}{\partial \boldsymbol{\theta}}(\boldsymbol{\theta}) + \frac{\partial^2 \Phi}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^T}(\boldsymbol{\theta})(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})$$

for $\boldsymbol{\theta}$ close to $\hat{\boldsymbol{\theta}}$. Rearranging, we have

$$\hat{\boldsymbol{\theta}} - \boldsymbol{\theta} \approx -\left\{ \frac{\partial^2 \Phi}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^T}(\boldsymbol{\theta}) \right\}^{-1} \frac{\partial \Phi}{\partial \boldsymbol{\theta}}(\boldsymbol{\theta})$$

This approximation suggests the following Newton-Raphson updating formula, given a starting value $\hat{\boldsymbol{\theta}}^0$.

$$\hat{\boldsymbol{\theta}}^{j+1} = \hat{\boldsymbol{\theta}}^j - \left\{ \frac{\partial^2 \Phi}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^T}(\hat{\boldsymbol{\theta}}^j) \right\}^{-1} \frac{\partial \Phi}{\partial \boldsymbol{\theta}}(\hat{\boldsymbol{\theta}}^j)$$

where $j = 1, 2, \ldots,$ convergence.

Through matrix differentiation, it is not difficult to obtain:

i. $\frac{\partial \Phi}{\partial \boldsymbol{\theta}}(\boldsymbol{\theta}) = -2\{\mathbf{J}(\boldsymbol{\theta})\}^T\{\mathbf{y} - \boldsymbol{\eta}(\boldsymbol{\theta})\}$;

ii. $\frac{\partial^2 \Phi}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^T}(\boldsymbol{\theta}) = 2\{\mathbf{J}(\boldsymbol{\theta})\}^T\{\mathbf{J}(\boldsymbol{\theta})\} - 2\frac{\partial \{\mathbf{J}(\boldsymbol{\theta})\}^T}{\partial \boldsymbol{\theta}}\{\mathbf{y} - \boldsymbol{\eta}(\boldsymbol{\theta})\}$

Therefore, the Newton-Raphson update formula in the context of the NLS is given by

$$\hat{\boldsymbol{\theta}}^{j+1} = \hat{\boldsymbol{\theta}}^j + \rho \cdot \left[\underline{\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}^T\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\} - \mathbf{D}^j}\right]^{-1} \cdot \{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}^T\{\mathbf{y} - \boldsymbol{\eta}(\hat{\boldsymbol{\theta}}^j)\} \qquad (42)$$

where $\mathbf{D}^j = \frac{\partial \{\mathbf{J}(\boldsymbol{\theta})\}^T}{\partial \boldsymbol{\theta}}\{\mathbf{y} - \boldsymbol{\eta}(\boldsymbol{\theta})\}\big|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}^j}$.

Recall that the Gauss-Newton update was

$$\hat{\boldsymbol{\theta}}^{j+1} = \hat{\boldsymbol{\theta}}^j + \rho \cdot \left[\underline{\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}^T\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}}\right]^{-1} \cdot \{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}^T\{\mathbf{y} - \boldsymbol{\eta}(\hat{\boldsymbol{\theta}}^j)\}$$

These two algorithms are obviously very similar. As compared with Newton-Raphson, the Gauss-Newton algorithm just replaces $\frac{\partial^2\Phi}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}^T}(\boldsymbol{\theta}) = 2\{\mathbf{J}(\boldsymbol{\theta})\}^T\{\mathbf{J}(\boldsymbol{\theta})\} - 2\mathbf{D}$ with $2\{\mathbf{J}(\boldsymbol{\theta})\}^T\{\mathbf{J}(\boldsymbol{\theta})\}$ (its expected value) and ignores $2\mathbf{D}$, in which the information from the second derivative is contained. Note that, in the N-R algorithm, $\boldsymbol{\Gamma} = \left[\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\}^T\{\mathbf{J}(\hat{\boldsymbol{\theta}}^j)\} - \mathbf{D}^j\right]^{-1}$.

2. *Quasi-Newton methods.* The N-R algorithm has appealing convergence properties, but computing the second derivatives may be excessively expensive making it difficult to implement in practice. Quasi-Newton methods avoid this by using numerical approximations to the second derivative matrix. Let

$$\mathbf{H}(\boldsymbol{\theta}) = \frac{\partial^2\Phi}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}^T}(\boldsymbol{\theta}) \quad \text{and} \quad \mathbf{g}(\boldsymbol{\theta}) = \frac{\partial\Phi}{\partial\boldsymbol{\theta}}(\boldsymbol{\theta})$$

$\mathbf{H}$ is also known as *Hessian matrix* and $\mathbf{g}$ is called *gradient matrix*.

In the N-R algorithm we had the updating formula

$$\hat{\boldsymbol{\theta}}^{j+1} = \hat{\boldsymbol{\theta}}^j - \{\mathbf{H}(\hat{\boldsymbol{\theta}}^j)\}^{-1}\mathbf{g}(\hat{\boldsymbol{\theta}}^j)$$

Also

$$\mathbf{g}(\hat{\boldsymbol{\theta}}^{j+1}) \approx \mathbf{g}(\hat{\boldsymbol{\theta}}^j) + \mathbf{H}(\hat{\boldsymbol{\theta}}^j)(\hat{\boldsymbol{\theta}}^{j+1} - \hat{\boldsymbol{\theta}}^j)$$

for $\hat{\boldsymbol{\theta}}^{j+1}$ close to $\hat{\boldsymbol{\theta}}^j$. Rearranging, we have

$$\mathbf{H}(\hat{\boldsymbol{\theta}}^j)(\hat{\boldsymbol{\theta}}^{j+1} - \hat{\boldsymbol{\theta}}^j) \approx \mathbf{g}(\hat{\boldsymbol{\theta}}^{j+1}) - \mathbf{g}(\hat{\boldsymbol{\theta}}^j)$$

with $\hat{\boldsymbol{\theta}}^{j+1} - \hat{\boldsymbol{\theta}}^j$ depends on $\mathbf{H}(\hat{\boldsymbol{\theta}}^j)$.

Therefore, in the Quasi-Newton methods, $\mathbf{H}^*(\boldsymbol{\theta})$ is used in place of $\mathbf{H}(\boldsymbol{\theta})$ in the N-R algorithm, where $\mathbf{H}^*(\boldsymbol{\theta})$ is chosen to satisfy Equation (43)

$$\mathbf{H}^*(\hat{\boldsymbol{\theta}}^{j+1})(\hat{\boldsymbol{\theta}}^{j+1} - \hat{\boldsymbol{\theta}}^j) = \mathbf{g}(\hat{\boldsymbol{\theta}}^{j+1}) - \mathbf{g}(\hat{\boldsymbol{\theta}}^j) \tag{43}$$

Initially, $\mathbf{H}^*(\hat{\boldsymbol{\theta}}^0)$ can be set to any symmetric positive definite matrix, for example, the identity matrix $\mathbf{I}$, then it is improved at each subsequent iteration by

$$\mathbf{H}^*(\hat{\boldsymbol{\theta}}^{j+1}) = \mathbf{H}^*(\hat{\boldsymbol{\theta}}^j) + \mathbf{E}^j$$

where $\mathbf{E}^j$ is an *update matrix* chosen so that (43) holds. A large number of Hessian updating methods have been developed. Generally, the formula of Broyden-Fletcher-Goldfarb-Shanno (BFGS) is thought to be the most effective for us in a general purpose method. Furthermore, to avoid the inversion

of the Hessian $\mathbf{H}(\boldsymbol{\theta})$, you can also derive an updating method in which the direct inversion of $\mathbf{H}(\boldsymbol{\theta})$ is avoid by using a formula that makes an approximation of the inverse Hessian $\mathbf{H}(\boldsymbol{\theta})^{-1}$ at each update. A well know procedure is the Davidon-Fletcher-Powell (DFP) formula. Note that the Q-N methods are appropriate for a problem which is not highly nonlinear but where the second derivatives are tedious to calculate. Compared with the G-N and N-R algorithms, the Q-N usually converges faster than the former but slower than the latter.

2.4 Uncertainty analysis

2.4.1 Confidence intervals for parameter and model prediction

The basic approach to conducting uncertainty analysis (i.e., providing confidence intervals) is to utilize the linear approximation to reduce the situation to the linear case and then use standard linear model results. Because of the approximation, these results will be approximate rather than exact. First of all, the variance of $\hat{\boldsymbol{\theta}}$ can be obtained by analogues with the linear model. That is,

$$\text{var}(\hat{\boldsymbol{\theta}}) \approx \sigma^2\{[\mathbf{J}(\boldsymbol{\theta}^*)]^T\mathbf{J}(\boldsymbol{\theta}^*)\}^{-1}$$

where $\boldsymbol{\theta}^*$ is the true value of $\boldsymbol{\theta}$. However, if we relax the assumption for the nonlinear model in (26) from $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2\mathbf{I}_n)$ to $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2\boldsymbol{\Sigma})$, $\text{var}(\hat{\boldsymbol{\theta}})$ should be obtained through the following formula

$$\text{var}(\hat{\boldsymbol{\theta}}) \approx \sigma^2\{[\mathbf{J}(\boldsymbol{\theta}^*)]^T\boldsymbol{\Sigma}^{-1}\mathbf{J}(\boldsymbol{\theta}^*)\}^{-1} \tag{44}$$

We estimate this asymptotic (approximate) variance-covariance matrix using

$$\widehat{\text{var}}(\hat{\boldsymbol{\theta}}) = s^2\{[\mathbf{J}(\hat{\boldsymbol{\theta}})]^T\mathbf{Q}\mathbf{J}(\hat{\boldsymbol{\theta}})\}^{-1} \tag{45}$$

where, $\mathbf{Q} = \boldsymbol{\Sigma}^{-1}$ as defined as in (17), and

$$s^2 = \frac{1}{n-p}\sum_{i=1}^{n}[w_i(y_i - f(\mathbf{x}_i, \hat{\boldsymbol{\theta}}))]^2 \tag{46}$$

Therefore, the confidence intervals for θ's and $E(y_0)$ can be obtained in an analogous fashion by replacing $\mathbf{X}^T\mathbf{Q}\mathbf{X}$ in the results for linear models with $\{[\mathbf{J}(\hat{\boldsymbol{\theta}})]^T\mathbf{Q}\mathbf{J}(\hat{\boldsymbol{\theta}})\}^{-1}$ in the nonlinear models. That is, the $100(1-\alpha)\%$ confidence interval for θ's is

$$\hat{\theta}_j \pm t_{\alpha/2}(n-p)\text{s.e.}(\hat{\theta}_j) \tag{47}$$

where

$$\text{s.e.}(\hat{\theta}_j) = s\sqrt{\{[\mathbf{J}(\hat{\boldsymbol{\theta}})]^T\mathbf{Q}\mathbf{J}(\hat{\boldsymbol{\theta}})\}^{-1}_{jj}}$$

The 95% confidence interval of each parameter provided by PEST is calculated using (47) when $\alpha = 0.05$.

Likewise, the linear approximation confidence interval for the expected response ($E(y_0)$) in nonlinear models can be generated using the analogs of the formula for the linear models, (23). We simply replace the estimated value $\mathbf{x}_0\hat{\boldsymbol{\beta}}$ by $f(\mathbf{x}_0, \hat{\boldsymbol{\theta}})$, the matrix $\mathbf{X}$ by $J(\hat{\boldsymbol{\theta}})$, and the vector $\mathbf{x}_0$ by

$$\mathbf{j}_0 = \frac{\partial f(\mathbf{x}_0, \hat{\boldsymbol{\theta}})}{\partial \boldsymbol{\theta}^T}$$

Then, the $100(1-\alpha)\%$ confidence interval for the expected response ($E(y_0)$) is (PEST does not provide this confidence interval.)

$$f(\mathbf{x}_0, \hat{\boldsymbol{\theta}}) \pm t_{\alpha/2}(n-p)\text{s.e.}(\hat{y_0}) \tag{48}$$

where

$$\text{s.e.}(\hat{y_0}) = s\sqrt{\mathbf{j}_0^T(\mathbf{J}(\hat{\boldsymbol{\theta}})^T\mathbf{Q}\mathbf{J}(\hat{\boldsymbol{\theta}}))^{-1}\mathbf{j}_0}.$$

It should be noted that the parameter correlation matrix provided by PEST is derived from the variance-covariance matrix (45). Each element in the resulting parameter correlation matrix is calculated using the following formula:

$$\rho_{ij} = \frac{\widehat{\text{cov}}(\beta_i, \beta_j)}{\sqrt{\widehat{\text{var}}(\beta_i)}\sqrt{\widehat{\text{var}}(\beta_j)}} \tag{49}$$

where, ρ_{ij} is the correlation coefficient of parameters β_i and β_j. It takes values between 0 and 1, with a value higher than 0.8 being considered highly (linearly) dependent on each other.

2.4.2 Nonlinear calibration-constrained method

This linear (or first-order) uncertainty analysis has the advantage that it can be implemented with virtually no computational burden. While the results of such an analysis can be useful for assessing parameter uncertainty in a *relative* sense, and ascertaining the degree of correlation between model parameters, its use in analyzing predictive uncertainty is often limited.

Through generalization of Equation (22), for a nonlinear model, a $100(1-\alpha)\%$ joint confidence region for all model parameters ($\boldsymbol{\theta}$) can be approximated by:

$$\Phi(\boldsymbol{\theta}) - \Phi(\hat{\boldsymbol{\theta}}) \leq ps^2 F_\alpha(p, n-p) \tag{50}$$

where $\Phi(\boldsymbol{\theta})$ is the objective function calculated for a parameter set $\boldsymbol{\theta}$ and $\Phi(\hat{\boldsymbol{\theta}})$ is the objective function calculated for the optimized (or estimated) parameter set $\hat{\boldsymbol{\theta}}$ (i.e., the minimized objective function obtained in the parameter estimation process), p, n, $F_\alpha(p, n-p)$, and s^2 have the same definition in (22). s^2, the sum of squared error, can be calculated using Equation (46) or simply by $s^2 = \Phi(\hat{\boldsymbol{\theta}})/(n-p)$. At this point, it is very important to make a distinction between inequalities (22) and (50). Both sides of (22) are quadratic functions of $\boldsymbol{\beta}$, confidence regions based on (22) are ellipsoids. But for (50), $\Phi(\boldsymbol{\theta}) - \Phi(\hat{\boldsymbol{\theta}})$ is only *approximately* equal to a quadratic function of $\boldsymbol{\theta}$. Using the linear approximation

$$\boldsymbol{\eta}(\boldsymbol{\theta}) \approx \boldsymbol{\eta}(\hat{\boldsymbol{\theta}}) + \mathbf{J}(\hat{\boldsymbol{\theta}})(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}) \tag{51}$$

where $\hat{\boldsymbol{\theta}}$ is the MLE (maximum likelihood estimator), we have

$$\begin{aligned} \Phi(\boldsymbol{\theta}) &= (\mathbf{y} - \boldsymbol{\eta}(\boldsymbol{\theta}))^T (\mathbf{y} - \boldsymbol{\eta}(\boldsymbol{\theta})) \\ &\approx (\mathbf{y} - \boldsymbol{\eta}(\hat{\boldsymbol{\theta}}))^T (\mathbf{y} - \boldsymbol{\eta}(\hat{\boldsymbol{\theta}})) - 2\underbrace{(\mathbf{y} - \boldsymbol{\eta}(\hat{\boldsymbol{\theta}}))^T \mathbf{J}(\hat{\boldsymbol{\theta}})}(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}) \\ &\quad + (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^T \mathbf{J}(\hat{\boldsymbol{\theta}})^T \mathbf{J}(\hat{\boldsymbol{\theta}})(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}) \end{aligned} \tag{52}$$

so that

$$\Phi(\boldsymbol{\theta}) - \Phi(\hat{\boldsymbol{\theta}}) \approx (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^T \mathbf{J}(\hat{\boldsymbol{\theta}})^T \mathbf{J}(\hat{\boldsymbol{\theta}})(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}) \tag{53}$$

Therefore, the linear approximation region of (50) takes the form,

$$\{\boldsymbol{\theta} | (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^T \mathbf{J}(\hat{\boldsymbol{\theta}})^T \mathbf{J}(\hat{\boldsymbol{\theta}})(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}) \leq ps^2 F_\alpha(p, n-p)\}.$$

Such regions are again ellipsoids, because $(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^T \mathbf{J}(\hat{\boldsymbol{\theta}})^T \mathbf{J}(\hat{\boldsymbol{\theta}})(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})$ is a quadratic function of $\boldsymbol{\theta}$. However, the approximation (53) can be very poor. The true contours of $\Phi(\boldsymbol{\theta}) - \Phi(\hat{\boldsymbol{\theta}})$ are not necessarily ellipses and ellipsoids. This disagreement can lead to poor estimates, see Figure 3 for two possible situations. When the number of observations (n) is large, this approximation is usually good. For a specific model prediction, a $100(1-\alpha)\%$ confidence interval can then be formed through minimizing/maximizing that prediction

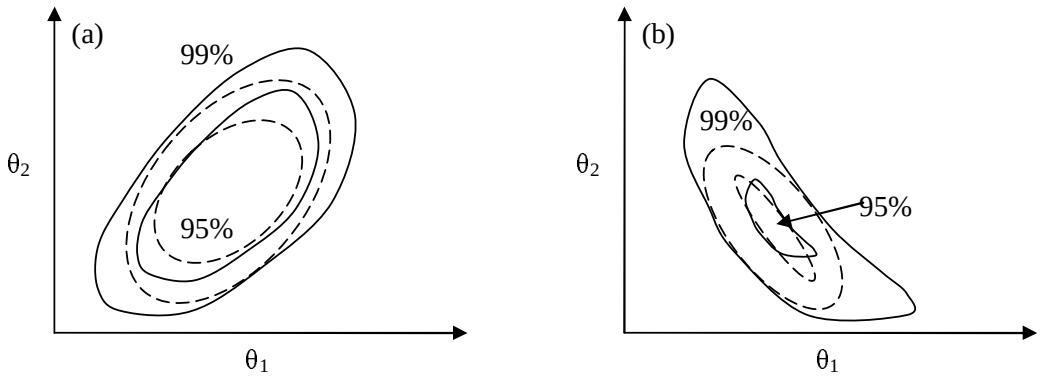


Figure 3: Confidence contour for $\boldsymbol{\theta} = (\theta_1, \theta_2)^T$.

while constraining the model parameters ($\boldsymbol{\theta}$) to lie within the joint confidence region defined by (50).

When applying this nonlinear calibration-constrained predictive uncertainty analysis method, we first specify a prediction of interest (for example, a 7-day average flow rate) whose uncertainty requires exploration. Then we find a parameter set that maximizes or minimizes that prediction while still maintaining the model in a calibrated state. The calibrated state is defined by an upper objective function limit, $\Phi_\alpha(\boldsymbol{\theta})$, a number that is slightly larger than the minimum objective function, $\Phi(\boldsymbol{\theta})$, obtained in the model calibration state. For example, to construct a 95% confidence interval for a 7-day average flow prediction, the calibrated state is calculated using the following equation which resulted from manipulating Equation (50).

$$\Phi_{0.05}(\boldsymbol{\theta}) = \Phi(\hat{\boldsymbol{\theta}}) + ps^2 F_{0.05}(p, n-p) \tag{54}$$

Care should be exercised in applying Equation (54) too rigorously for its validity is based on the assumption that the n observations used for model calibration are independent from each other, which is rarely the case in environmental modeling. Hence, a more qualitative assessment of $\Phi_\alpha(\boldsymbol{\theta})$ may be warranted, perhaps based on visual inspection of model-to-measurement misfit. Also note that it is the maximum and the minimum values of the specified model prediction, instead of the parameter sets generating these extreme values, that is our major interest. Therefore, the prediction confidence interval limited by the maximum and minimum values respectively represent the upper and lower limits of the simultaneous confidence interval

of the specified prediction.

3 Miscellaneous

3.1 Convergence criteria

Let $\hat{\boldsymbol{\theta}}^k$ denote the estimate from the kth iteration. Convergence can be measured in several ways:

1. Look at change in the parameter estimates:

   If $\hat{\boldsymbol{\theta}}^k \approx \hat{\boldsymbol{\theta}}^{k-1}$ we may consider the sequence $\hat{\boldsymbol{\theta}}^0, \hat{\boldsymbol{\theta}}^1, \hat{\boldsymbol{\theta}}^2, \ldots$ to have converged and take $\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}^k$ as the NLS estimator. The most common way to establish that $\hat{\boldsymbol{\theta}}^k \approx \hat{\boldsymbol{\theta}}^{k-1}$ is to look at

   $$\max_j \frac{\left|\hat{\theta}_j^k - \hat{\theta}_j^{k-1}\right|}{\left|\hat{\theta}_j^{k-1}\right|}.$$

   If this criterion is less than some tolerance value (e.g., 1×10^{-6}) the algorithm stops.

2. Look at change in the objective function $S(\boldsymbol{\theta})$:

   If $\Phi(\hat{\boldsymbol{\theta}}^k) \approx \Phi(\hat{\boldsymbol{\theta}}^{k-1})$ then take $\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}^k$. To judge this we can use the criterion

   $$\frac{\Phi(\hat{\boldsymbol{\theta}}^{k-1}) - \Phi(\hat{\boldsymbol{\theta}}^k)}{\Phi(\hat{\boldsymbol{\theta}}^k)}.$$

   If this criterion is smaller than some tolerance value, then the algorithm stops.

   Which criterion should be used depends on the estimation purpose. If the main interest is in the mean response, $\boldsymbol{\eta}(\boldsymbol{\theta})$, then (2) may be more appropriate. If the model's parameters themselves are of more interest then (1) perhaps is better.

3. Use the normal equation (31): Both the previous criteria have the disadvantage that they do not indicate when $\hat{\boldsymbol{\theta}}^k$ is close to $\hat{\boldsymbol{\theta}}$ or when $\Phi(\hat{\boldsymbol{\theta}}^k)$ is close to $\Phi(\hat{\boldsymbol{\theta}})$. For example, if we knew the minimum value of the least squares criterion $\Phi(\hat{\boldsymbol{\theta}})$ was equal to certain value (say, 2.56), then we could stop the algorithm based on judging when $\Phi(\hat{\boldsymbol{\theta}}^k) \approx$

2.56. However, we would never know the values of $\Phi(\hat{\boldsymbol{\theta}})$ or $\hat{\boldsymbol{\theta}}$ to judge convergence of the algorithm. But we do know that $\hat{\boldsymbol{\theta}}$ must satisfy the normal equation

$$[\mathbf{y} - \boldsymbol{\eta}(\hat{\boldsymbol{\theta}})]^T \mathbf{V}(\hat{\boldsymbol{\theta}}) = \mathbf{0}$$

Therefore, we can stop the algorithms when[3]

$$[\mathbf{y} - \boldsymbol{\eta}(\hat{\boldsymbol{\theta}}^k)]^T \mathbf{V}(\hat{\boldsymbol{\theta}}^k) \approx \mathbf{0} \tag{55}$$

3.2 Derivatives computation

As shown above, all the numerical methods utilize the derivative matrix $\mathbf{J}(\boldsymbol{\theta}) = \partial f(\mathbf{x}, \boldsymbol{\theta})/\partial \boldsymbol{\theta}^T$. These routines usually use numerical derivatives rather than analytic derivatives since the latter is not attainable in most cases. However, the numerical derivatives can be inaccurate and slow to compute, especially when second derivatives are needed such as in the N-R algorithm. Therefore, accuracy in derivatives calculation is fundamental to these method's success in optimizing parameters. The most common cause of the failure to converge to the global minimum is the presence of roundoff errors incurred in the calculation of derivatives. Three numerical differentiation schemes were used to calculate the derivatives – forward-difference, backward-difference, and central-difference methods. The previous two methods take less computation time but are less accurate than the third method. Therefore, in PEST, the faster forward-difference scheme is used to calculate the numerical derivatives when $\Phi(\hat{\boldsymbol{\theta}}^j)$ is far away from the minimum of $\Phi(\hat{\boldsymbol{\theta}})$, while the more accurate central-difference scheme is used when $\Phi(\hat{\boldsymbol{\theta}}^j)$ is close to the minimum.

3.3 Parameter estimation of compartmental models

Environmental systems are usually described by compartmental models, that is, they are mathematically represented by systems of (ordinary or partial) differential equations. The relationships between dependent and explanatory variables are not explicitly described as in the nonlinear regression models,

[3]Equation (55) itself is not normally used as a convergence criterion in parameter estimation exercises, but does suggest the third criterion, sometimes called "Relative Offset Criterion" superior to the previous two. The interested readers can refer to the reference for more details.

for which, all the numerical methods mentioned above are used in parameter estimation. In order to estimate the parameters of a compartmental model described in a system of ordinary differential equations, one of the following two approaches can be used.

1. If the system of ordinary differential equations (ODE) is simple and can be solved analytically, then a nonlinear regression-like model can be obtained to describe the relationship between the dependent and explanatory variables. Then parameter estimation methods can be directly applied to this regression model, a solution of the system of ODEs.

2. If the system is complex and the system of ODEs has no closed form solutions, then numerical integration has to be resorted to solve the ODEs and numerical finite-difference methods are used to calculate the derivatives.

3.4 Initial values and prior information

For the numerical methods mentioned in Section 2.3 and other commonly used algorithms for fitting nonlinear models, initial values of the parameters are needed. Bad initial values can lead to slow or no convergence, or convergence to a local minimum of the objective function. For environmental models, the initial values of the parameters usually come from two sources: experiment and literature. But there is usually no guarantee that the initial values you selected for the parameters will lead to convergence. Therefore, it is suggested that before an iterative method is used to automatically search for the optimal values for parameters, a set of reasonable parameter values should be obtained manually (trial and error) and set to be the initial values.

It can also happen that we have some prior information concerning the parameters that we wish to optimize. It is often useful to include this information in the parameter estimation process since it may lend stability to the process, diminish the non-uniqueness of the parameter estimates, etc. The ways to including the prior information into the estimation process are various, such as setting upper and lower bounds to the parameters, reparameterization, adding rows containing the prior information to the model matrix, or introducing a "penalty function" to the objective function.

3.5 Parameter transformation

Parameter transformation has two common purposes: (1) to enforce constraints (prior information) on parameter as mentioned above; (2) to facilitate algorithm convergence. Logarithm, centering and scaling are three commonly used parameter transformation methods and they can be specified by the users when using PEST.

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